

EARTH SYSTEM TREND DETECTIVE

EarthTrend Detective Pro

Investigate how Earth-system changes unfold.

TEMPERATURE

PRECIPITATION

WIND

SPATIAL EVIDENCE

TREND DNA

What changed here?

Heat



Rain



Wind



Spatial consistency



**Evidence first.
Interpretation second.**

Earth is changing. The hard part is proving the pattern.

A trend detective needs more than a rising line.

WHAT?

Which Earth-system variable changed?

Temperature
Precipitation
Wind

WHERE?

Is the signal local or spatially consistent?

Point evidence
5 x 5 sweep

HOW MUCH?

What is the direction and rate of change?

Theil-Sen slope
Calendar-year gaps

IS IT SIGNIFICANT?

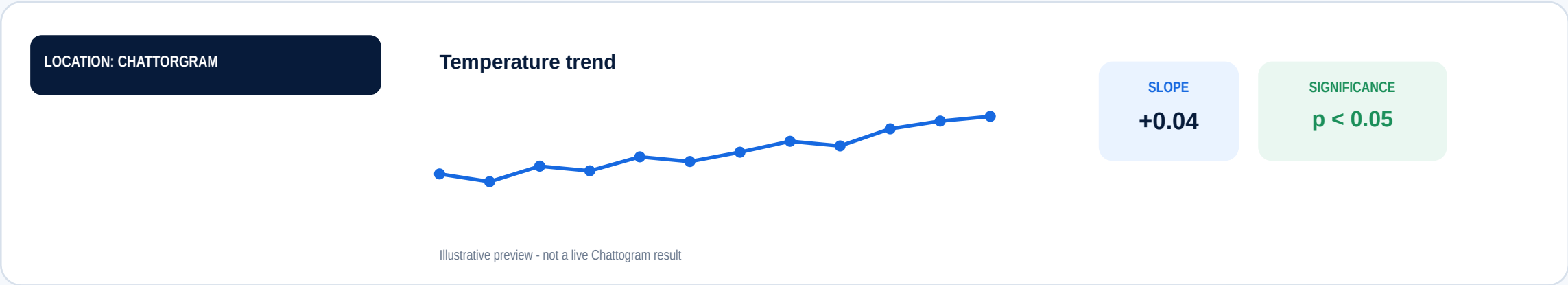
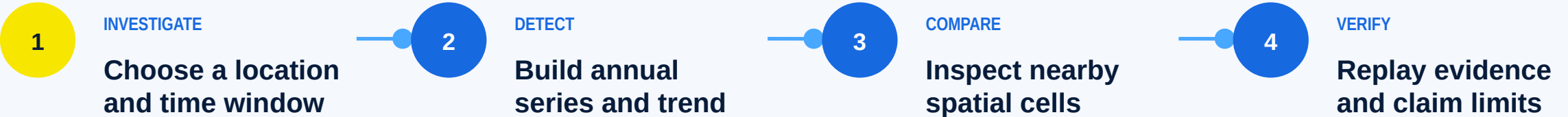
Could the pattern be noise or autocorrelation?

Modified Mann-Kendall
FDR correction

Challenge focus: detect change over time, measure magnitude, compare locations, and test whether the trend is statistically significant.

Turn raw Earth data into an investigation.

A clear workflow from location to evidence.



One interface. One evidence chain. No black-box verdict.

Four moves. One investigation.

Designed for fast exploration without sacrificing scientific context.

01

SELECT

Location + time

Chattogram, BD

02

ANALYZE

Trend + significance

Sen slope / MK



03

MAP

Spatial pattern

5 x 5 neighborhood



04

REPLAY

Evidence + limits

Source / method / claim

- Source
- Quality
- Baseline
- Trend
- Significance

Designed to be readable on a laptop, projector, and phone.

NASA data is the engine, not decoration.

The analysis starts with NASA POWER and preserves the evidence trail.

NASA POWER

NASA

T2M

Temperature

PRECTOTCORR

Precipitation

WS10M

Wind

Monthly inputs -> annual series -> trend statistics

EVIDENCE CHAIN

- Source
- Quality gate
- Annualization
- Baseline
- Trend
- Significance
- Spatial correction**

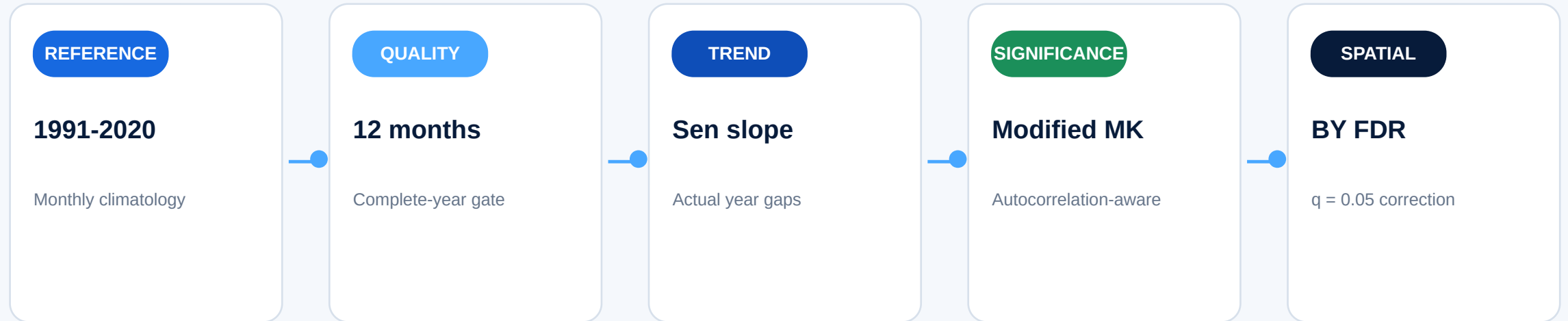
CLAIM GUARDRAILS

Trend != causality
Point != regional average
Non-significance != no change

Source: NASA POWER documentation

A trend is not a claim. Evidence comes first.

Built to expose the assumptions behind every headline number.



INTERPRETATION RULES

Significant trend = statistical evidence of non-random change. NOT causality.

Short records are descriptive only; no significance claim.

From trend detection to Earth-system investigation.

A reusable evidence workflow for real environmental questions.

HEAT STRESS

Detect persistent warming signals

Temperature

WATER + RAINFALL

Inspect changing precipitation patterns

Precipitation

ENVIRONMENTAL STRESS

Compare trend fingerprints across space

Spatial evidence

FINAL PROJECT

EarthTrend Detective Pro

<https://earth-trend-detective.base44.app/>

SOURCE CODE

GitHub repository: add your public repo URL before submission

TRY THE APP



Built by decoder | NASA Space Apps Challenge 2026